Single-tier Aggregation

1. What is Single-tier Aggregation?

Single-tier aggregation means using one layer of servers to receive raw events, combine them in memory, and periodically write summarized results to the database.

Instead of writing every event:

```None
A

B

C

B

C

D
```

Write:

```None
A = 1

B = 5

C = 7

D = 6
```

This greatly reduces database writes.

2. How It Works

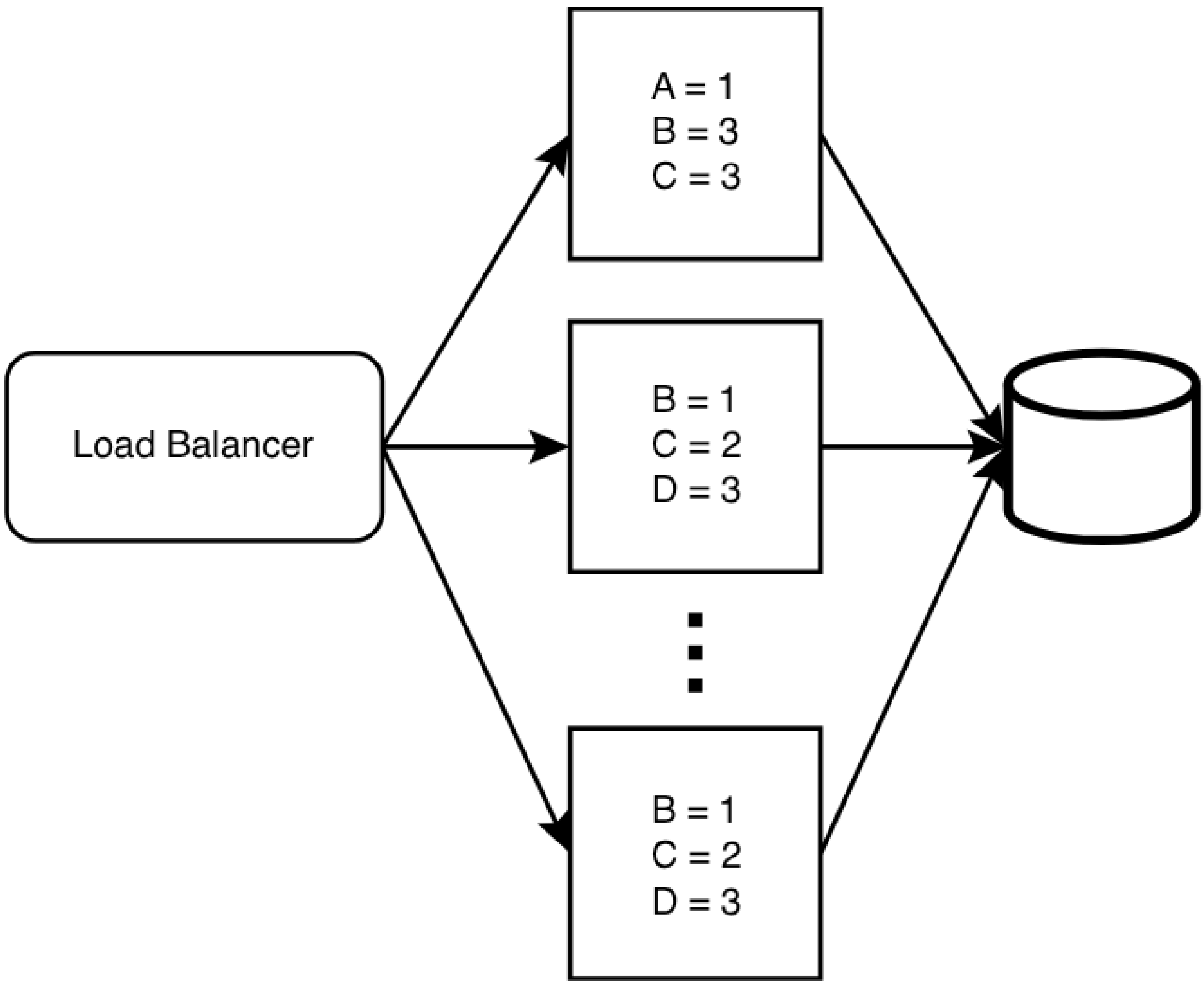

Figure shows a **load balancer** distributing events evenly across multiple hosts.

Each host stores counts in an in-memory hash table.

Example:

```None
Host 1:

A = 1

B = 3

C = 3
```

```None
Host 2:

B = 1

C = 2

D = 3
```

```None
Host 3:

B = 1

C = 2

D = 3
```

3. Final Combined Output

When flushed to database:

```None
A = 1

B = 5

C = 7

D = 6
```

4. Why It Matters

Without aggregation:

```None
Every event = 1 database write
```

With aggregation:

```None
Thousands of events = few summary writes
```

This reduces:

```None
Write load

Disk I/O

Replication traffic

Storage growth

Database scaling cost
```

5. Memory Structure Used

Each host usually uses:

```None
HashMap<Key, Count>
```

Example:

```None
{

 A:1,

 B:3,

 C:3

}
```

Fast updates in RAM.

6. Flush Strategy

Hosts write data to DB:

```None
Every 5 minutes

OR

When memory full

OR

At threshold count
```

7. Real World Example

Website page views:

```None
home clicked

home clicked

product clicked
```

Stored as:

```None
home = 2

product = 1
```

8. Advantages

```None
Very high write reduction

Fast in-memory counting

Cheap scaling

Simple architecture

Better performance
```

9. Disadvantages

```None
Per-event details lost

Crash before flush may lose counts

Delayed final database updates

Needs merging logic
```

10. Fault Tolerance Needed

To avoid count loss:

```None
Replication

Checkpointing

Kafka replay

Periodic snapshots
```

11. When to Use

Best for:

- Click counts
- Views
- Search frequency
- Sensor counts
- Likes/reactions
- API metrics

12. When Not to Use

Avoid when each event matters:

```None
Payments

Audit logs

Bank transactions

Security logs
```

13. Interview Talking Point

If asked to reduce DB writes:

```None
Use single-tier aggregation layer

Count in memory

Flush periodically

Store final totals only
```

Strong HLD answer.

14. One-Line Summary

Single-tier aggregation uses one layer of servers to combine incoming events in memory and periodically store summarized counts, reducing database writes and improving scalability.